An AI-Powered Eye-Tracking using MediaPipe and Laser Pointing Interface for Paralyzed Patients

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***Abstract*— The use of assistive communication systems with eye-gaze tracking has been of prime significance in recent times, as it provides a vital interface for severely motor-impaired patients, such as those suffering from Amyotrophic Lateral Sclerosis (ALS) and Locked-In Syndrome. This paper introduces a low-cost, real-time laser-based eye-gaze detection system using computer vision and embedded systems. The suggested approach utilizes MediaPipe's Face Mesh API to obtain dense facial and iris feature points from a regular RGB camera. The gaze is detected by calculating the normalized iris position relative to ocular feature points, whereas confirmation of user commands is performed by detecting blinks using Eye Aspect Ratio (EAR) values. The detected gaze coordinates are then transmitted to a microcontroller, which operates a 2-DOF pan-tilt servo mechanism with a laser pointer. The efficacy of the suggested approach is validated by conducting experiments to demonstrate its ability to deliver accurate gaze classification and rapid-speed physical pointing in regular lighting conditions.**

Keywords— MediaPipe, Amyotrophic Lateral Sclerosis (ALS), Eye Aspect Ratio (EAR)

I. Introduction

Human-computer interaction has traditionally involved physical devices such as keyboards, mouse, and touch screens. However, there are people suffering from severe motor disorders such as Amyotrophic Lateral Sclerosis (ALS), spinal cord injuries, and Locked-In Syndrome. They are unable to control any voluntary muscles and hence are unable to interact in the usual manner. The only reliable way for such people to communicate is through eye movements. Eye gaze tracking technology enables us to track where the user is looking by monitoring the movements of the user's eyes. Recent advances in computer vision and artificial intelligence allow us to build such systems at a very low cost using regular cameras [1].

A. Motivation

However, most of the available eye tracking systems are expensive and require specialized hardware, which is a drawback for their use in resource-constrained environments. Additionally, most of these systems are limited to screen-based interaction and are not capable of interacting with physical devices in any way. Hence, there is a need for a portable, cost-effective, and non-intrusive gaze-based communication system that is capable of translating eye movement to real-world actions.

B. Objective of the Work

The objective of this project is to design and implement an AI-based laser eye gaze detection system that can be used to perform hands-free interaction. The system can track the gaze direction and perform intentional blinking, which can be mapped to physical interaction using the pan-tilt laser mechanism. Motivated by the rising necessity to incorporate assistive technologies into the fields of healthcare and human-computer interaction, eye gaze tracking has evolved as an influential technology that can be used to enable communication among physically impaired individuals. Especially when there is severe paralysis due to stroke and Amyotrophic Lateral Sclerosis (ALS), eye gaze tracking can be used as a non-invasive interaction mechanism. The following section discusses the existing research in the fields of eye gaze tracking, iris detection, blink interaction, embedded control systems, and low-cost assistive technologies.

II. Literature Review

A. MediaPipe- Based Eye Gaze Tracking Systems

Significant advancements in real-time facial landmark detection have been achieved in recent times due to the introduction of MediaPipe by Google Research. MediaPipe Face Mesh is capable of detecting facial and iris landmark detection with high precision by using lightweight machine learning models for real-time applications. Several studies have been conducted to show that MediaPipe-based systems exhibit high robustness in moderate lighting conditions and are capable of accurate gaze estimation without any special infrared equipment installed [2]. It was also demonstrated that iris-based landmark tracking enhances depth estimation and directional accuracy for real-time human-computer interaction systems. However, some limitations were observed in extreme lighting changes, head movement, and occlusion caused by eyeglasses

B. Iris Localization and Gaze Estimation Techniques

Traditional gaze estimation methods relied on pupil detection using thresholding and edge detection techniques [3]. While computationally simple, these approaches were highly sensitive to illumination changes. More recent works incorporate machine learning models and geometric mapping methods for estimating gaze direction. Regression-based mapping between iris centre displacement and screen coordinates has shown improved accuracy. Hybrid methods combining geometric eye modelling and neural network-based correction have demonstrated enhanced precision, especially in dynamic environments. Despite improvements, many systems remain limited to screen-based interaction and do not extend gaze estimation to physical control mechanisms.

C. Blink Detection Using Eye Aspect Ratio (EAR)

In the traditional method of gaze estimation, the detection of the pupil was carried out through thresholding and edge detection. Though this method was computationally simple, it was highly susceptible to changes in illumination. In recent times, the use of machine learning and geometric mapping has been implemented in gaze estimation [4]. Regression-based mapping has shown promising results in the field. A combination of geometric eye modelling and neural network has shown good precision in the field. Though there are many advancements in the field of gaze estimation, many systems are limited to screen-based control and do not extend the estimation to physical control systems.

D. Embedded Systems for Assistive Control

Low-cost embedded systems like ESP32 and Arduino have been successfully implemented in assistive and robotic control applications. These embedded systems are capable of wireless communication, servo control, and real-time response. Literature on the integration of embedded systems with vision systems emphasizes the need for minimizing communication delays between the processing unit and the hardware actuators. Serial communication has been successfully implemented for reliable communication between Python-based vision systems and microcontrollers [5]. However, issues like synchronization delays and power optimization are still areas of active research

E. Pan–Tilt Mechanisms and Laser-Based Assistive Interfaces

Servo-actuated pan-tilt mechanisms are often utilized in robotic vision and tracking applications [6]. Studies have shown that 2-DOF servo configurations can provide accurate direction control with minimal hardware complexity. Laser pointer-based feedback mechanisms have been studied in rehabilitation and interactive applications for visual feedback of user intent. These types of feedback mechanisms need to consider eye safety and control of laser beam intensity. Most gaze trackers are designed to focus on virtual cursor control and do not consider translating gaze into physical space using mechanical actuators [7].

III. Mathematical Model

A. Coordinate Conversion

MediaPipe provides normalized coordinates (x_n, y_n). These are converted to pixel coordinates using

$$x = x_n W \quad (1)$$

$$y = y_n H \quad (2)$$

where W and H denote the width and height of the camera frame respectively [8].

B. Iris Center Estimation

The iris centre is calculated as the average of the detected iris landmark points

$$x_{iris} = \frac{1}{N}\sum_{i=1}^{N} x_i \quad (3)$$

$$y_{iris} = \frac{1}{N}\sum_{i=1}^{N} y_i \quad (4)$$

where $N =$ represents the number of iris landmarks [3].

C. Euclidean Distance

The distance between two landmark points is computed using the Euclidean distance formula

$$d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]} \quad (5)$$

This metric is used to compute eye width and height [3],[9].

D. Eye Geometry Calculation

Eye Width

$$W_{eye} = \sqrt{[(x_r - x_l)^2 + (y_r - y_l)^2]} \quad (6)$$

Eye Height

$$H_{eye} = \sqrt{[(x_b - x_t)^2 + (y_b - y_t)^2]} \quad (7)$$

E. Gaze Ratio Estimation

Horizontal Gaze Ratio

$$G_x = \frac{x_{iris} - x_l}{W_{eye}} \quad (8)$$

Vertical Gaze Ratio

$$G_y = \frac{y_{iris} - y_t}{H_{eye}} \quad (9)$$

These normalized ratios determine the gaze direction independent of camera distance [1],[2].

F. Direction Classification

The gaze direction is determined by thresholding the gaze ratios

$$G_x < T_L \rightarrow LEFT \quad (10)$$
$$G_x > T_R \rightarrow RIGHT \quad (11)$$
$$G_y < T_U \rightarrow UP \quad (12)$$
$$G_y > T_D \rightarrow DOWN \quad (13)$$

Otherwise,

$$CENTER \quad (14)$$

where T_L, T_R, T_U, T_D represent threshold values.

G. Eye Aspect Ratio for Blink Detection

$$EAR = \frac{\| P_2 - P_6 \| + \| P_3 - P_5 \|}{2 \| P_1 - P_4 \|} \quad (15)$$

If $EAR < T_{blink}$ (16)
a blink event is detected [10],[12]

H. Servo Angle Update

Servo motion is updated incrementally based on the gaze command [6],[7].

$$\theta_{new} = \theta_{old} \pm \Delta\theta \quad (17)$$

where $\Delta\theta$ represents the step angle.

The servo angle is constrained by

$$0^\circ \leq \theta \leq 180^\circ \quad (18)$$

I. Laser Point Projection

If the target board is located at distance L, the laser point position can be approximated as

$$X = L\tan(\theta_{pan}) \quad (19)$$

$$Y = L\tan(\theta_{tilt}) \quad (20)$$

where θ_{pan} and θ_{tilt} denote the pan and tilt servo angles.

IV. METHODOLOGY

The proposed system will be a continuous closed-loop pipelined system. It will include image acquisition, facial landmark detection, gaze estimation, blink detection, and actuation. It will be used to convert the movement of the iris into physical laser pointing.

A. Image Acquisition

In the proposed system, ESP32-CAM or any webcam is used to capture the user's face images in real-time. The captured video frames are generally in RGB colour format, which is the native colour format of OpenCV. Sometimes, it may be in grayscale format also. To make it suitable for MediaPipe's processing pipeline, it is converted to RGB colour format. Additionally, mirroring of the images is performed to make it natural for user interaction

B. Facial Landmark Detection

The system uses MediaPipe Face Mesh to identify 468 landmarks on the face. Of these, the landmarks of the eye corners, eyelids, and iris are used to perform the gaze detection and blink detection [11]. This method ensures that the performance of the system is not affected by small head movements and changes in illumination conditions

C. Iris Center Estimation

The centre of the iris is found by finding the centroid of the identified iris landmarks. This is used as a reference point to represent the gaze direction in the eye region. Depending on the position of the reference point, the movement of the eye is found to determine whether the gaze is towards the left or the right.

D. Gaze Direction Estimation

The horizontal and vertical gaze ratios are computed based on the extent to which the iris position changes within the boundaries of the eye under normal conditions. The gaze direction is then classified as left, right, up, down, and centre using predefined threshold values

E. Blink detection

Blink detection is achieved through the Eye Aspect Ratio (EAR), which is defined as the ratio of the sum of vertical distances between eyelid landmarks and twice the horizontal distance between eye corner landmarks. Intentional blinks are detected when the EAR is below a certain threshold for a certain period of time and are then used as a selection command [12].

F. Decision Logic

The control command is determined based on the combination of the direction of the gaze and the duration of the blink, thereby avoiding any unintended command during natural eye movements. For example, during a left gaze, a long blink detected over a specific time interval indicates a selection command.

G. Embedded Actuation

The processed command is then transmitted to the microcontroller. The microcontroller then controls the pan and tilt motors as well as the laser pointer. Moreover, the user's requirements are shown on an OLED display.

H. System floor representation

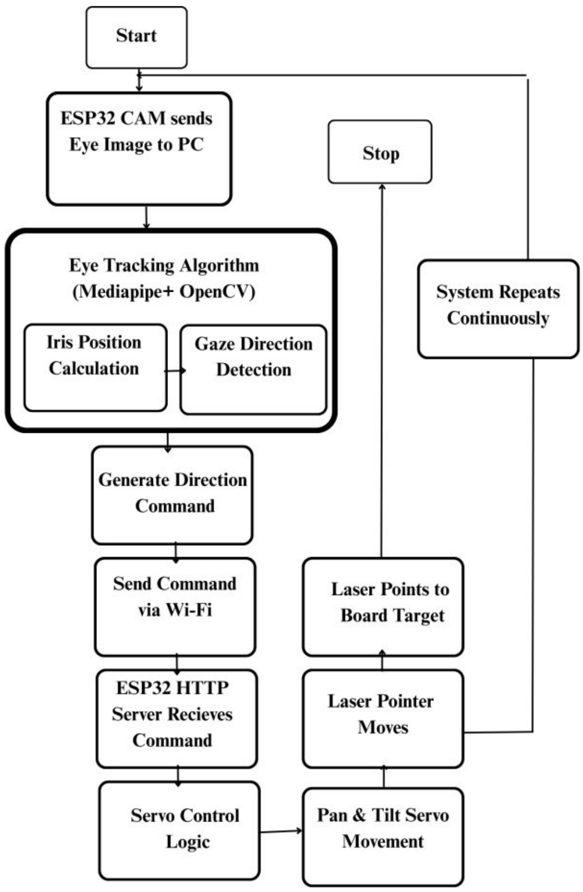


Fig. 1. Overall system flowchart

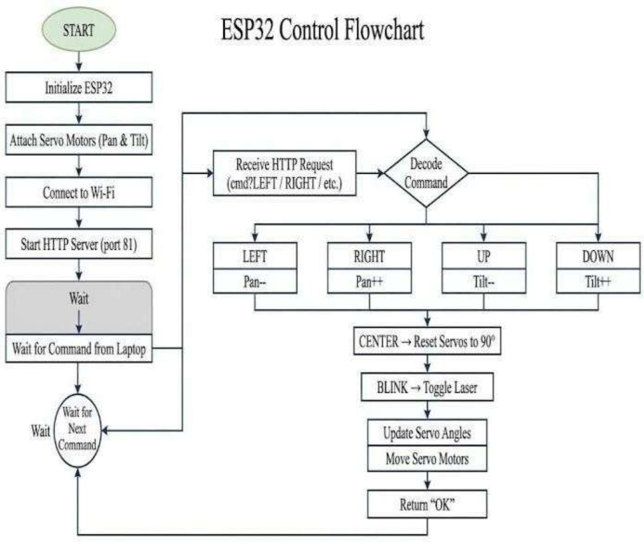


Fig. 2. ESP32 control flowchart

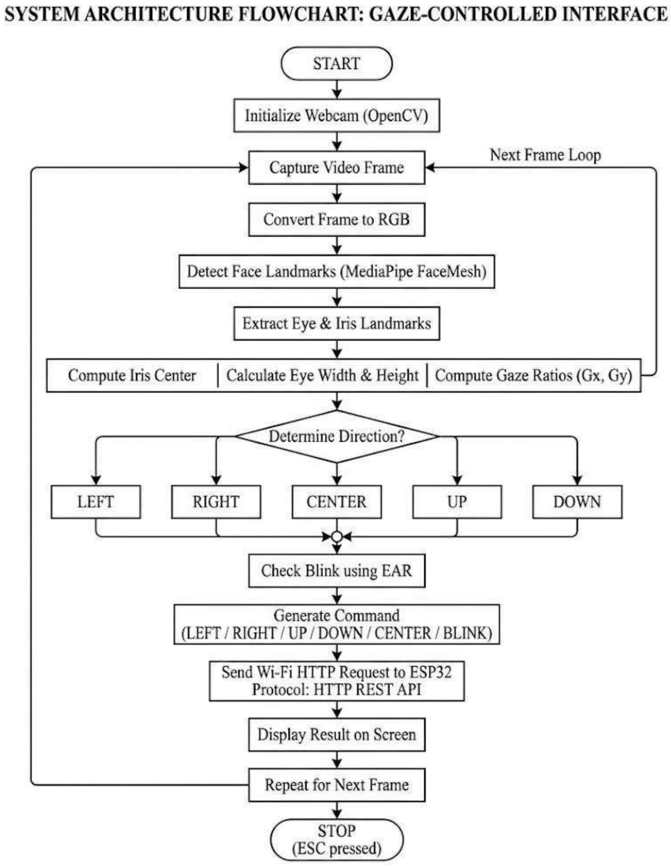


Fig. 3. Gaze-control

V. RESULT AND DISCUSSION

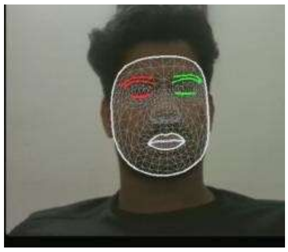
The proposed system was implemented using the MediaPipe framework to enable real-time face and eye tracking for assistive interaction. Experimental results demonstrate that the system successfully detects and interprets user intent through eye movements and blinking patterns.

Initially, the system detects and localizes the user's face using MediaPipe's face detection module. Once the face is identified, the facial landmark model extracts key points, particularly focusing on the eye region. The system then detects the iris and computes its center point in real time. To interpret gaze direction, the system partitions the eye region into multiple zones (left, right, up, and down). By continuously tracking the relative position of the iris center within these partitions, the system determines the direction of eye movement. Experimental observations confirm that this partition-based approach provides reliable gaze estimation under varying lighting conditions and moderate head movements.

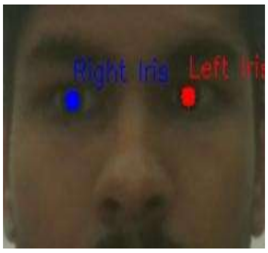
Furthermore, the system incorporates a long-blink detection mechanism [e.g., > 500 ms] to act as a selection or confirmation signal. When the user gazes toward a particular direction and performs a prolonged blink, the system interprets it as an intentional command. For instance, a rightward gaze combined with a long blink signal that the user wants the option on the right.

Integrating gaze direction tracking with blink-based confirmation significantly reduces false activations and enhances interaction accuracy. Results indicate that the proposed approach is effective for assistive communication, particularly for users with limited motor abilities, enabling them to express needs through simple eye movements and controlled blinking.

(a) 468 land marks Detected



(b) Face Mesh



(c) Iris Detection

(d) Gaze Detection

Fig. 4. Eye Tracking Algorithm Outputs

VI. CONCLUSION

This project demonstrates a low-cost, real-time system that converts eye movements into physical pointing using computer vision and embedded control. By using MediaPipe Face Mesh for facial landmark detection and simple gaze normalization, the system can reliably determine eye direction from a standard RGB camera. Blink detection using the Eye Aspect Ratio (EAR) allows users to confirm commands naturally. The detected gaze direction is sent to an ESP32 microcontroller, which controls a pan–tilt servo mechanism with a laser pointer. Experimental results show that the system works effectively under normal indoor lighting and provides responsive interaction using affordable hardware. Overall, the proposed system shows strong potential as an accessible assistive tool for individuals with limited motor abilities

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